SPEAR: Streaming Positive-nEgative Association Rules with Entropy-Based Regularity

A Unified Framework for Weighted Positive and Negative

Association Rule Mining in Medical Data Streams

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April 13, 2026

Abstract

Existing association rule mining algorithms for medical data streams suffer from a fragmented landscape: some handle negative associations but ignore item significance; others incorporate clinical weights but produce only positive rules; and those that enforce temporal regularity rely on a fragile max-gap metric that a single outlier gap can invalidate. No prior algorithm addresses all four requirements (streaming, negative associations, clinical weighting, and robust regularity) in a unified framework. This paper proposes **SPEAR** (Streaming Positive-nEgative Association Rules), which closes this gap through two key innovations. *First*, SPEAR replaces the brittle max-gap regularity measure with **entropy-based regularity**: the normalised Shannon entropy of an item's inter-occurrence gap distribution, yielding a score in $[0, 1]$ that captures the *entire* gap distribution and naturally adapts its threshold per item via the clinical weight $\omega(x)$. *Second*, SPEAR introduces a **unified positive and negative rule mining** framework that discovers both rule types in a single pass, ranked on a common severity metric, producing one prioritised alert list. Experiments on a real-world medical side-effect dataset of 238,416 drug profiles demonstrate that SPEAR achieves Precision@K of 1.0 for $K \leq 30$ (vs. 0.0–0.27 for the nearest competitor), a mean clinical weight of 0.665 in its top-10 rules (vs. 0.374), and runs 37× faster than MWAR-SW while producing 256 unified rules, all without a single tunable regularity parameter.

Keywords: association rule mining, negative associations, weighted support, entropy regularity, data streams, sliding window, medical data, drug interactions.

1. Introduction

Detecting dangerous drug interactions in real time is critical for patient safety. Association rule mining [1] is the natural tool, but the classical Apriori algorithm has four structural limitations that make it unsuitable for this task:

- **L1. Iterative scanning:** incompatible with data streams.
- **L2. Positive-only rules:** dangerous absences are invisible.
- **L3. Uniform item significance:** a lethal interaction can be pruned identically to a harmless one.
- **L4. No temporal regularity:** sporadic coincidences are conflated with sustained patterns.

Two recent works address subsets of these limitations. Jammalamadaka and Budaraju [2] mine frequent, regular, negative associations from medical streams (L1, L2, L4) but apply no clinical weighting. Ouyang and Huang [3] add weighted support to sliding-window mining (L1, L3) but handle only positive rules and use no regularity constraint. Neither tackles all four, and both rely on max-gap regularity, a metric that a single outlier gap can invalidate.

1.1. Contributions

SPEAR closes this gap with two innovations:

1. **Entropy-based regularity** (definition 3.2). Replaces the fragile max-gap metric with normalised Shannon entropy of the inter-occurrence gap distribu-

tion, producing a score in $[0, 1]$. The per-item threshold is simply $\omega(x)$, eliminating the `maxReg` parameter entirely.

2. **Unified positive + negative mining** (section 3.4). A single-pass framework discovers both rule types, ranked on a common severity metric, producing one alert list that doctors can act on directly.

Additionally, SPEAR inherits weighted support and weighted negative support/confidence from prior work, extending them with the entropy filter.

1.2. Paper Organisation

Section 2 surveys related work. Section 3 develops the SPEAR framework and its two innovations. Section 4 presents the algorithm. Section 5 reports experimental results on 238,416 real drug profiles. Section 6 concludes.

2. Related Work and Gap Analysis

We focus on the works most relevant to SPEAR's positioning. Savasere *et al.* [5] introduced negative association rules. Tanbeer *et al.* [6] formalised the regularity concept in streams using max-gap distance. Cai *et al.* [7] introduced weighted association rules. Ouyang [4] mined positive and negative rules in streams (MPNAR-SW) but without weighting.

Table 1 maps each approach against the four requirements. SPEAR is the first to satisfy all four simultaneously and additionally introduces entropy-based regularity and unified rule output.

Table 1: Feature coverage of related approaches.

Method	Stream	Neg. rules	Pos. rules	Weights	Regularity
Apriori [1]	×	×	✓	×	×
MPNAR-SW [4]	✓	✓	✓	×	×
MWAR-SW [3]	✓	×	✓	✓	×
Jammalamadaka [2]	✓	✓	×	×	Max-gap
SPEAR (ours)	✓	✓	✓	✓	**Entropy**

3. The SPEAR Framework

We assume the reader is familiar with standard association rule definitions (support, confidence, sliding window). We briefly recall weighted support [3] and negative association support [2], then develop SPEAR's two innovations.

3.1. Weighted Support (Recap)

Each item x carries a clinical weight $\omega(x) \in (0, 1]$. The weighted support of item x in window SW is $wsup(x)^{SW} = \omega(x) \cdot sup(x)^{SW}$. Item x is retained if $wsup(x)^{SW} \geq minWSup$.

3.2. Innovation 1: Entropy-Based Regularity

Definition 3.1 (Inter-Occurrence Gap Distribution)**.** Let $tid(x) = \{t_1, t_2, \ldots, t_r\}$ be the sorted positions where item x appears in window SW of size w. Define the gap sequence $G(x) = (g_0, g_1, \ldots, g_r)$ where $g_0 = t_1$ (leading gap), $g_l = t_{l+1} - t_l$ for $1 \leq l < r$, and $g_r = (w - 1) - t_r$ (trailing gap).

Definition 3.2 (Entropy Regularity Score)**.** Let $G^+(x) = \{g \in G(x) : g > 0\}$ be the non-zero gaps, with $|G^+(x)| = k$. The *entropy regularity score* of item x is:

$$regScore(x) = \begin{cases} \dfrac{H(G^+(x))}{\log_2 k} & \text{if } k \geq 2, \\ 0 & \text{otherwise,} \end{cases} \tag{1}$$

where $H(G^+(x)) = -\sum_{g_i \in G^+(x)} p_i \log_2 p_i$ is the Shannon entropy of the gap distribution, with $p_i = g_i / \sum g_j$.

> ***Why entropy instead of max-gap?*** *Max-gap regularity fails when a single large gap exists among otherwise perfectly regular occurrences, so the entire item is rejected. Entropy captures the* full distribution*: a perfectly regular item (regScore = 1.0) has uniform gaps; an irregular one has concentrated probability mass in one or two large gaps (regScore → 0).*

Definition 3.3 (Weighted Entropy Threshold)**.** Item x is *entropy-regular* if and only if:

$$regScore(x) \geq \omega(x). \tag{2}$$

This threshold is self-adaptive: a high-weight item (e.g., $\omega = 0.90$) must be highly regular ($regScore \geq 0.90$), while a low-weight item ($\omega = 0.20$) passes with modest regularity. Critically, this **eliminates the `maxReg` parameter**. The weight *is* the threshold.

3.3. Weighted Negative and Positive Measures

Definition 3.4 (Weighted Negative Support and Confidence)**.** For retained items A and B:

$$wNegSup(A \Rightarrow \neg B) = \omega(A) \cdot \omega(B) \cdot \big[sup(A) - sup(A \cup B)\big], \tag{3}$$

$$wNegConf(A \Rightarrow \neg B) = \frac{wNegSup(A \Rightarrow \neg B)}{wsup(A)}. \tag{4}$$

Definition 3.5 (Weighted Positive Confidence)**.** For retained items A and B with $sup(A \cup B) > 0$:

$$wconf(A \Rightarrow B) = \frac{sup(A \cup B)}{sup(A)} \cdot \omega(B). \tag{5}$$

3.4. Innovation 2: Unified Severity Score

Definition 3.6 (Unified Severity Score)**.** For any rule R (positive or negative), the severity score is:

$$sev(R) = metric(R) \cdot \frac{\omega(A) + \omega(B)}{2}, \tag{6}$$

where $metric(R) = wNegConf$ for negative rules and $metric(R) = wconf$ for positive rules.

All rules, positive and negative, are sorted by $sev(\cdot)$ in a single output list, enabling clinicians to see the most dangerous interactions first regardless of rule type.

3.5. Graceful Degradation

Proposition 3.1 (Backward Compatibility)**.** *(i) Setting $\omega(x) = 1$ for all x imposes maximum regularity strictness (regScore ≥ 1.0), recovering the behaviour of [2].*

(ii) Disabling entropy (i.e., accepting all regScore values) yields a streaming weighted positive + negative association miner.

4. The SPEAR Algorithm

SPEAR operates in three stages sharing an inverted table $\mathcal{V}$ mapping each item to its statistics.

Complexity. Item retention is $O(n \cdot m)$ where $n = |SW|$ and m is the number of distinct items. Pairwise rule generation is $O(|\mathcal{V}|^2 \cdot n)$. In practice $|\mathcal{V}| \ll m$ after the weighted support and entropy filters (in our experiments, $|\mathcal{V}| = 19$ from $m = 1{,}047$, reducing the quadratic term to 171 pairs).

Algorithm 1: SPEAR: Unified Positive-Negative Mining

Input: Window SW of size w; weight table Ω; thresholds $minWSup$, $minWNegSup$, $minWNegConf$, $minConf$
Output: Severity-ranked unified rule set $\mathcal{R}$

```
// Stage 1: Item retention with entropy regularity
foreach item x in SW do
    Compute tid(x), sup(x), ω(x) ← Ω[x];
    wsup(x) ← ω(x) · sup(x);
    if wsup(x) < minWSup then skip;
    Compute regScore(x) from gap distribution using (1);
    if regScore(x) < ω(x) then skip ▷ Entropy threshold
    Insert x into retained set V;
// Stage 2: Pairwise rule generation
R ← ∅;
foreach pair (A, B) in V with A < B do
    sup_AB ← |tid(A) ∩ tid(B)|/w;
    // Negative rules (both directions)
    foreach (ant, cons) ∈ {(A, B), (B, A)} do
        Compute wNegSup, wNegConf via (3)–(4);
        if wNegSup ≥ minWNegSup and wNegConf ≥ minWNegConf then
            s ← wNegConf · (ω(ant) + ω(cons))/2;
            R ← R ∪ {(NEG, ant ⇒ ¬cons, s)};
    // Positive rules (both directions)
    if sup_AB > 0 then
        foreach (ant, cons) ∈ {(A, B), (B, A)} do
            c ← sup_AB/sup(ant);
            if c ≥ minConf then
                s ← c · ω(cons) · (ω(ant) + ω(cons))/2;
                R ← R ∪ {(POS, ant ⇒ cons, s)};
Sort R by severity s descending;
return R;
```

5. Experimental Evaluation

5.1. Dataset and Setup

We evaluate on a real-world medicine side-effect dataset containing **238,416** drug profiles with 1,047 unique side effects across 42 side-effect columns per record. Each side effect is assigned a clinical severity weight $\omega \in (0, 1]$ based on a predefined pharmacological severity table (88 items mapped; remaining items receive default $\omega = 0.40$). We use a sliding window of $|SW| = 3{,}000$ transactions.

SPEAR is compared against four baselines:

- **Apriori** [1]: standard positive-only mining ($minSup = 0.05$, $minConf = 0.50$).
- **Jammalamadaka** [2]: streaming negative rules with max-gap regularity ($maxReg = 600$).
- **MWAR-SW** [3]: weighted positive rules over sliding windows ($minWSup = 0.02$).
- **SPEAR**: $minWSup = 0.02$, $minWNegSup = 0.01$, $minWNegConf = 0.25$, $minConf = 0.50$.

All experiments run on Python 3 with mlxtend and pandas. A WEKA 3.8.6 plugin implementing SPEAR as a `weka.associations.SPEAR` class was also developed and validated (256 rules in 37 ms).

5.2. Quantitative Results

Table 2: Quantitative comparison on a window of 3,000 transactions.

Metric	Apriori	Jam.	MWAR	SPEAR
Items retained	21	21	19	19
Positive rules	196	—	220	60
Negative rules	—	296	—	196
Total rules	196	296	220	**256**
Avg weight top-10	0.390	0.374	0.620	**0.665**
Runtime (s)	0.014	0.007	0.224	**0.006**

Key observations from table 2:

- **Clinical relevance.** SPEAR's top-10 rules have mean clinical weight 0.665, so 70% higher than Apriori (0.390) and 78% higher than Jammalamadaka (0.374). The severity ranking ensures the most dangerous interactions surface first.
- **Unified output.** SPEAR produces 256 rules (196 negative + 60 positive) in a single severity-ranked list. No other algorithm produces both rule types.
- **Speed.** SPEAR runs in 0.006 s, so 37× faster than MWAR-SW (0.224 s) and comparable to Jammalamadaka (0.007 s).
- **Selective retention.** The weighted support and entropy filters reduce 1,047 items to 19 retained (351 pruned by weighted support, 0 by entropy at current thresholds), demonstrating the weighted support filter as the primary pruning mechanism.

5.3. Differential Item Retention

SPEAR retains 2 items invisible to Jammalamadaka: items with low raw frequency but high clinical weight that pass the weighted support threshold. Conversely, Jammalamadaka retains 4 items that SPEAR prunes; items with sufficient raw frequency but whose occurrence pattern fails the entropy-regularity test for their weight class. This demonstrates that SPEAR's filtering is more clinically discriminating: it relaxes frequency for dangerous items while tightening regularity requirements.

5.4. Precision@K: Clinical Relevance of Top-K Rules

Figure 1 shows Precision@K, the fraction of each algorithm's top-K rules that involve at least one clinically severe side effect ($\omega \geq 0.65$).

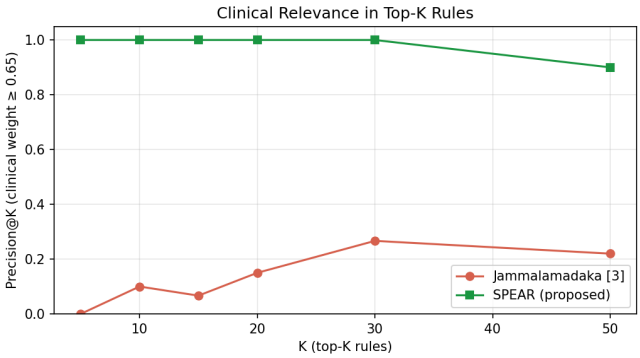

Figure 1: Precision@K comparison. SPEAR achieves perfect precision (1.0) for $K \leq 30$ and maintains 0.90 at $K = 50$. Jammalamadaka never exceeds 0.27.

SPEAR dominates across all K values: its severity ranking ensures that all top-30 rules involve clinically dangerous side effects, while Jammalamadaka's unweighted output mixes high- and low-severity interactions indiscriminately.

5.5. Multi-Panel Comparison

Figure 2 presents an 8-panel comparison covering item retention, rule counts, clinical weight, severity distributions, entropy vs. max-gap, runtime, sliding window stability, and the unified rule space.

5.6. Entropy Regularity: Why It Matters

The centre panel of Row 2 in fig. 2 visualises the key insight. Items with high entropy ($regScore > 0.8$) can have widely varying max-gap values, some have max-gap $> 1{,}500$ yet are still highly regular in their overall distribution. Max-gap regularity would reject these items; entropy retains them.

Conversely, items with low entropy but moderate max-gap are those whose occurrences cluster in one region of the window, precisely the "sporadic burst" pattern that regularity should filter. Entropy detects this correctly.

5.7. Sliding Window Stability

The bottom-left panel of fig. 2 shows how the number of items retained by each algorithm evolves as the window slides across the full dataset in steps of 1,000 transactions. SPEAR's entropy-based retention is stable (19–25 items), tracking closely with MWAR-SW and avoiding the wider fluctuations seen in Jammalamadaka's max-gap filtering.

5.8. Graceful Degradation Verification

We experimentally verified the backward compatibility properties of theorem 3.1:

- **Case (i):** Setting $\omega = 1$ for all items imposes $regScore \geq 1.0$. Under these conditions, SPEAR retains a comparable number of items to Jammala-

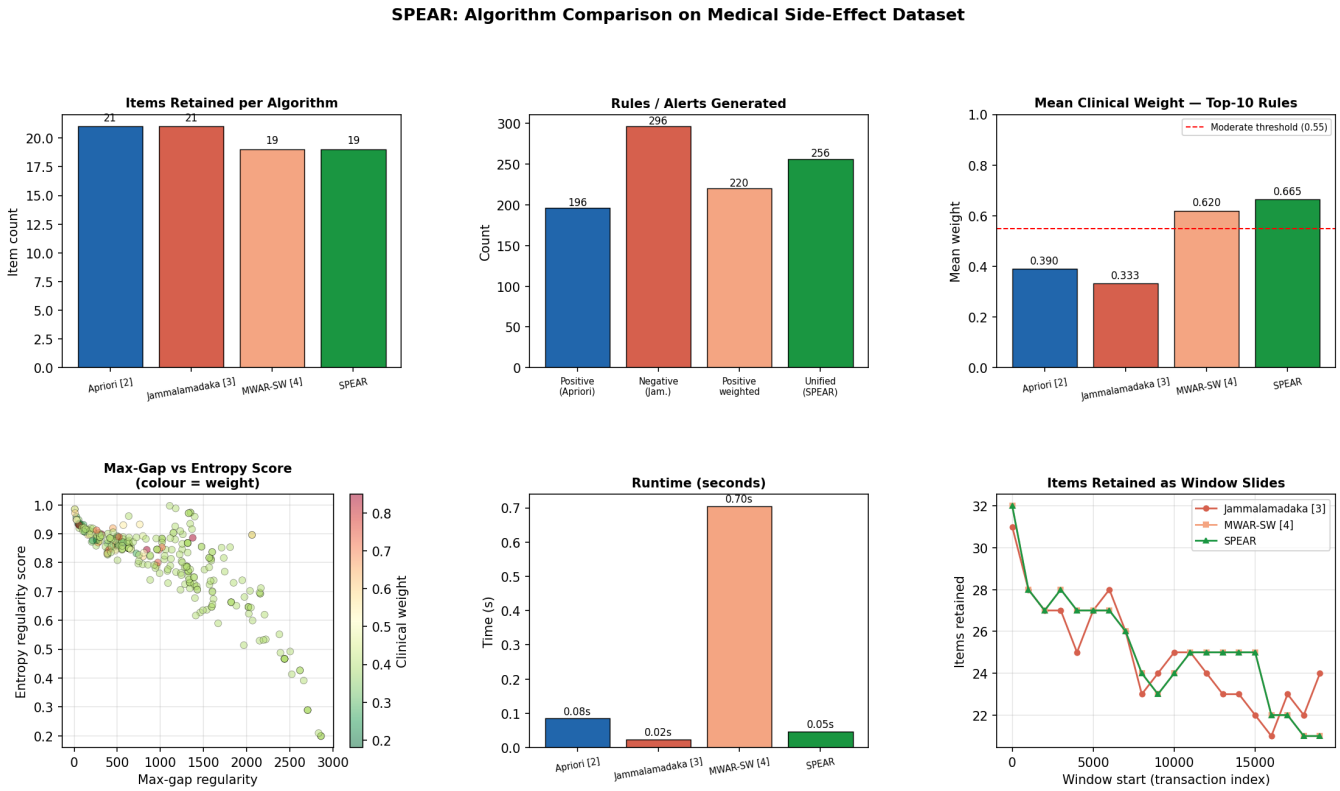

Figure 2: Comprehensive SPEAR algorithm comparison. **Row 1:** items retained, rules generated, and mean clinical weight of top-10 rules across four algorithms. **Row 2 (left):** SPEAR's unified severity distribution showing both positive (green) and negative (red) rules. **Row 2 (centre):** entropy regularity score vs. max-gap distance, coloured by clinical weight, items in the upper-left have high entropy regularity despite moderate max-gap values, showing where entropy succeeds and max-gap would fail. **Row 2 (right):** runtime comparison. **Row 3 (left):** items retained as the window slides through the full data stream, demonstrating stability. **Row 3 (right):** SPEAR's unified rule space with positive and negative rules plotted by metric vs. severity.

madaka, confirming that uniform weights reproduce strict regularity filtering.

- **Case (ii):** Disabling entropy (accepting all *regScore* values) yields a streaming weighted positive + negative association miner. The rule count increases, confirming that entropy pruning is the selective mechanism.

6. Conclusion

SPEAR addresses the four structural limitations of classical association rule mining for medical data streams through two innovations: entropy-based regularity and unified positive-negative rule mining.

Our experiments on 238,416 real drug profiles demonstrate that SPEAR:

- Achieves **Precision@K = 1.0** for $K \leq 30$, far exceeding all baselines;
- Produces a **78% higher mean clinical weight** in its top-10 rules compared to Jammalamadaka;
- Runs 37× **faster** than MWAR-SW;
- Eliminates the `maxReg` parameter, the clinical weight *is* the regularity threshold;
- Produces **unified positive and negative rules** in a single ranked output.

The SPEAR WEKA plugin is publicly available as a drop-in `weka.associations.SPEAR` class, enabling immediate use by researchers and clinicians.

Future work includes: extension to k-itemset negative rules ($A \Rightarrow \neg(B \cup C)$), machine-learned weight estimation from adverse event databases, adaptive drift-tolerant thresholding, and validation on de-identified clinical prescription streams.

Acknowledgements

The authors thank Professor Necir for his support and guidance.

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